

The background image shows a dense forest of tall evergreen trees. In the foreground, there is a fire burning on the ground, with smoke rising from it. The smoke is a hazy, orange-brown color, partially obscuring the trees in the background. The overall scene is dramatic and intense, with a strong sense of the power of a wildfire.

WILDFIRE SMOKE DISPERSION

Group 1: Justin Aranda, Ben Fan, Ryan Ng, and Ethan Wong

Objective

What factors contribute to the dispersion of wildfire smoke?

Specifically investigating smoke area's relationship with fire and weather parameters

Data Sources

- NOAA Hazard Mapping System
 - Fire and Smoke Data
- Open-Meteo Historical Weather API

Data Description & Management

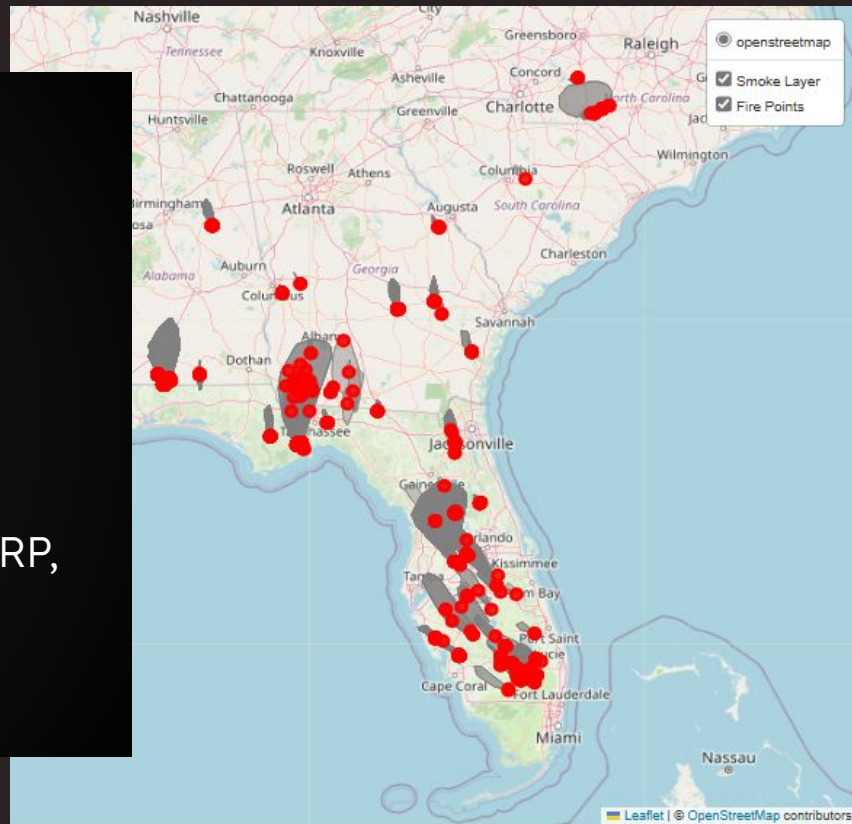
Data Description: HMS

Smoke NOAA HMS Data

- Geometry(Polygon), Density, Longitude, Latitude, Satellite, Method
- Aggregated: area_km^2

Fire NOAA HMS Data

- Geometry(Point), Longitude, Latitude, FRP, Ecosystem, Satellite, Method

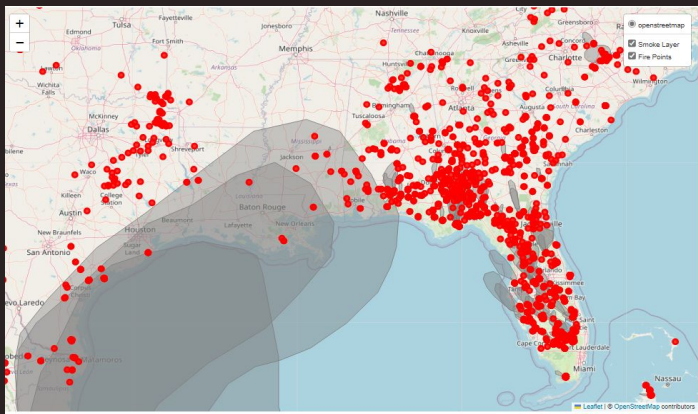


Data Management: HMS

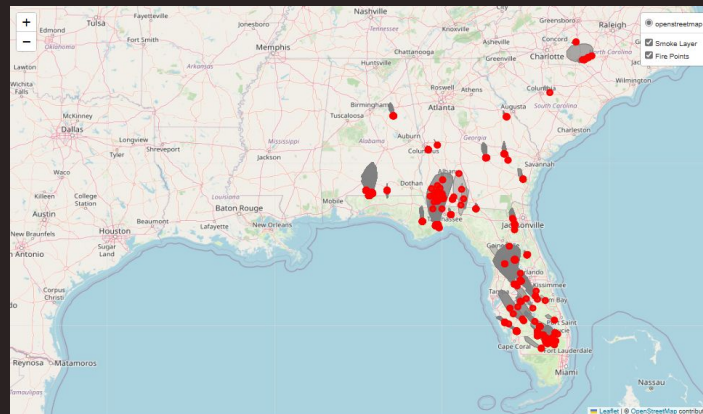
Fire and Smoke Sanity Check

- Filtered data given its within USA and territories
- Filtered out smoke which did not contain a fire data point... and vice versa using spatial-temporal joins

Before: 2/14/2023



After: 2/14/2023



Database Workflow



Data Description: Weather

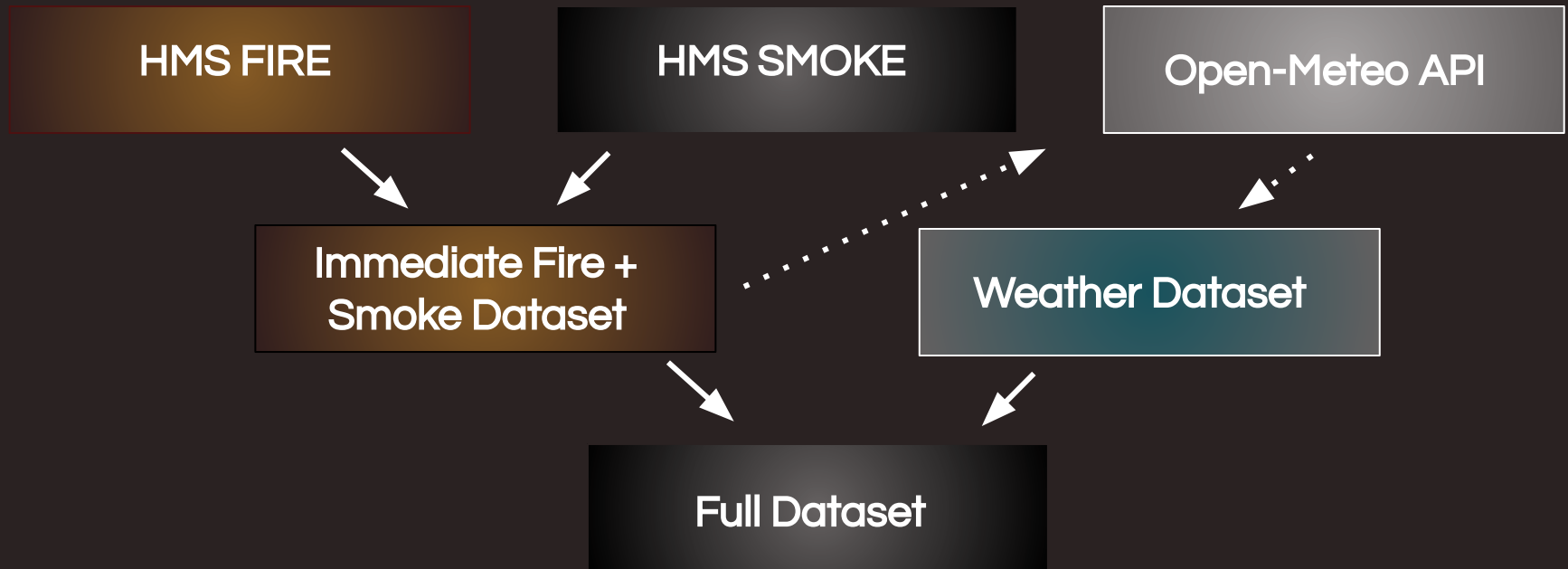
Temporal and Spatial Parameters

- Longitude, Latitude, Datetime_hour, Date

Weather Parameters

- **Temperature:** 'temperature_2m', 'temperature_2m_max', 'temperature_2m_min'
- **Wind:** 'wind_speed_10m', 'wind_direction_10m', 'wind_gusts_10m', 'wind_speed_10m_max', 'wind_gusts_10m_max', 'wind_direction_10m_dominant'
- **Misc:** 'soil_moisture_0_to_7cm', 'relative_humidity_2m', 'dew_point_2m', 'precipitation', 'precipitation_sum', 'rain_sum', 'shortwave_radiation_sum', 'sunshine_duration', 'daylight_duration'

Database Workflow



Database Tools

PYTHON

Used for API calls,
web scraping and
accessing
geospatial data

Libraries Used:
Geopandas, pandas,
numpy

R

Used for regression
analysis and data
visualization

Utilized Lasso and
LM regression,
ggplot2

POSTGRESQL

Querying to clean,
join, and verify data

Utilized PostGIS to
extend PostgreSQL
capabilities to work
with geospatial data

Stat/ML Methods

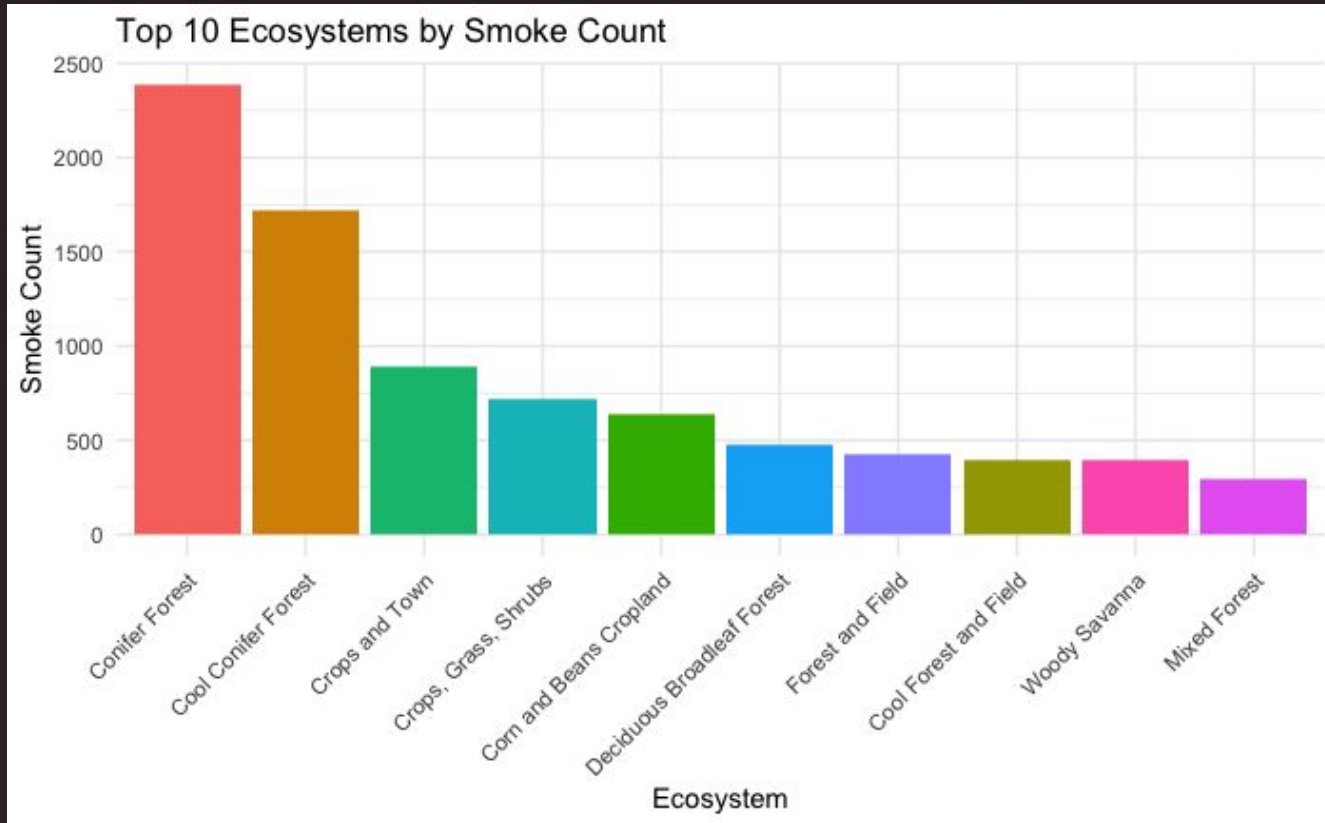
Methods Used

For the method testing, we used Linear and LASSO Regression & Random Forests

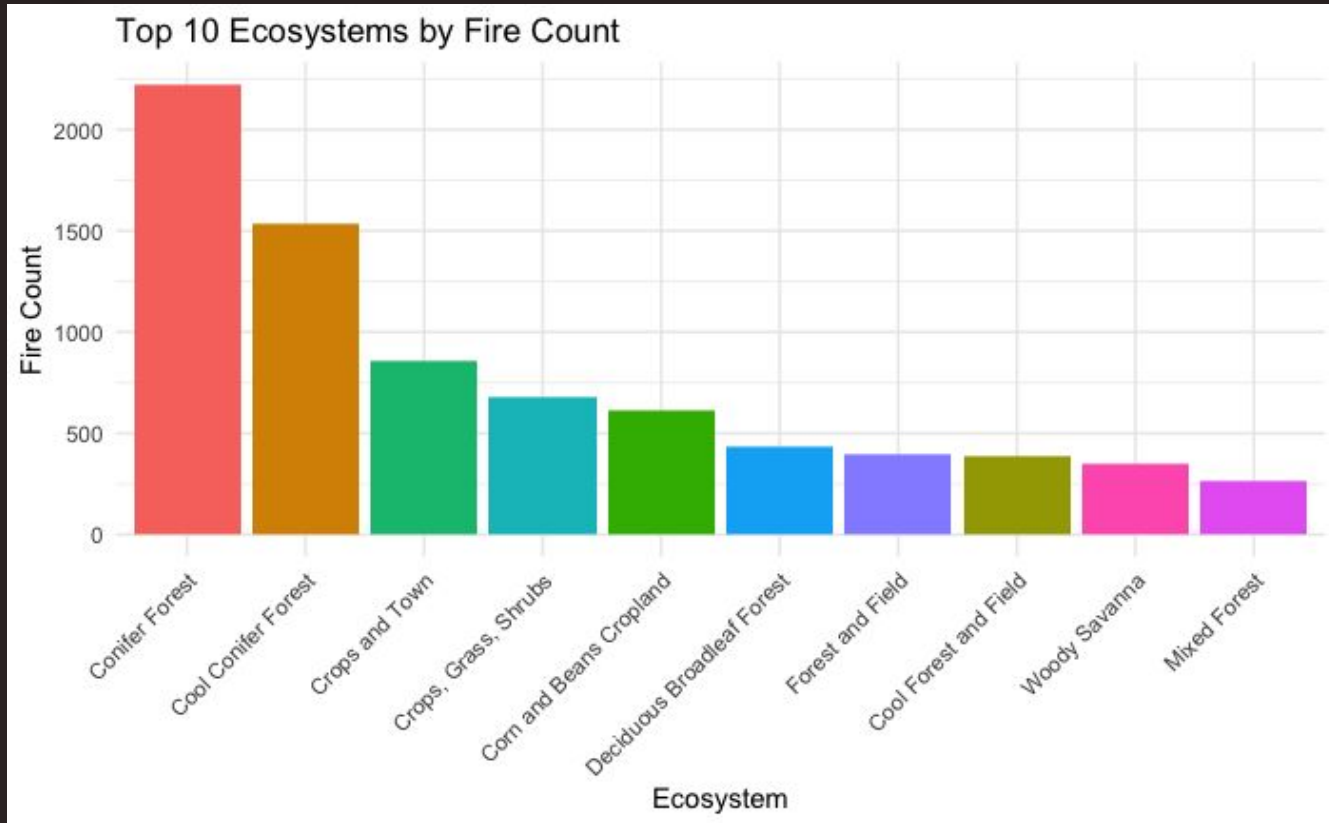
- Linear Regression: To find the smoke area coverage's relationship with other variables
- LASSO Regression: To test which interaction term has the highest correlation towards the area coverage
- Random Forests covered later in this presentation

Data Visualization

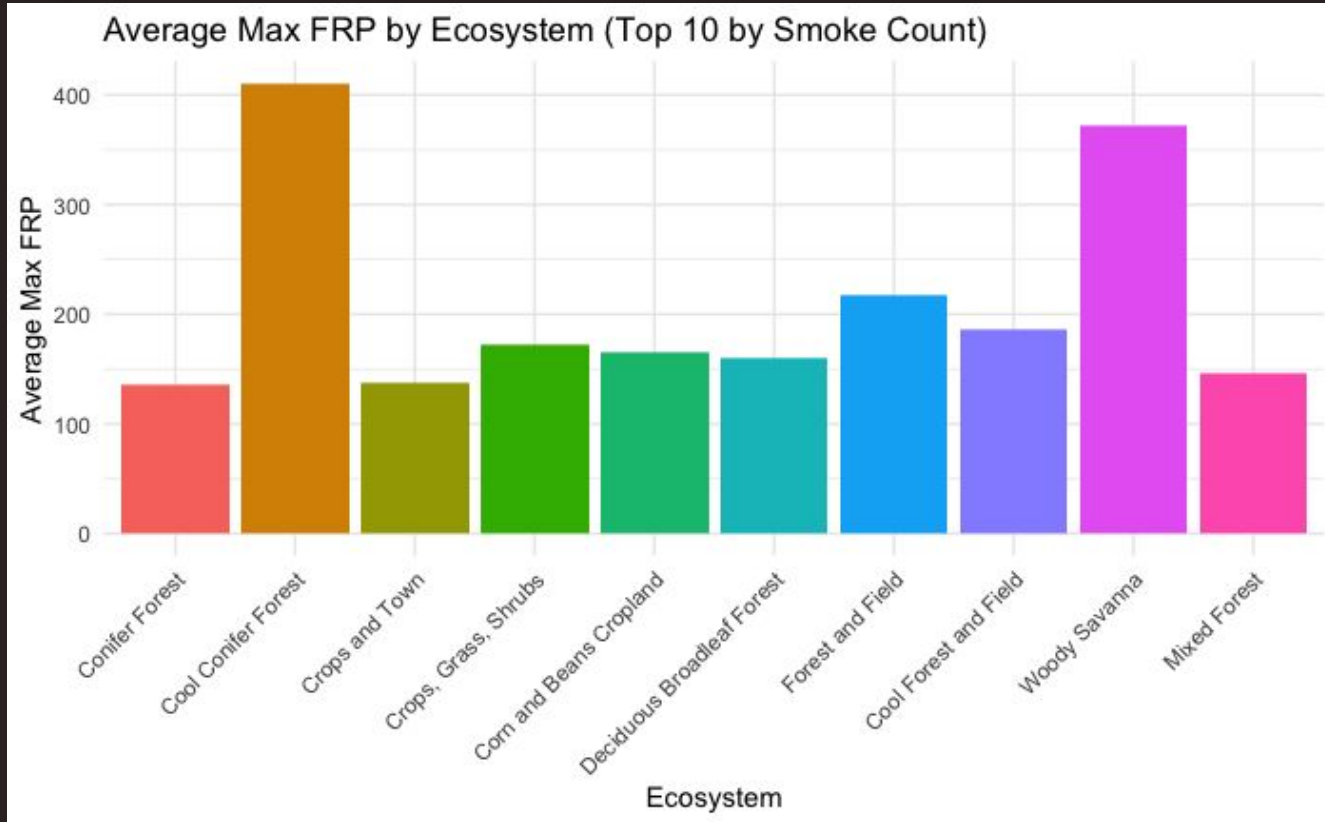
Ecosystems and Fire/Smoke Occurrences



Ecosystems and Fire/Smoke Occurrences

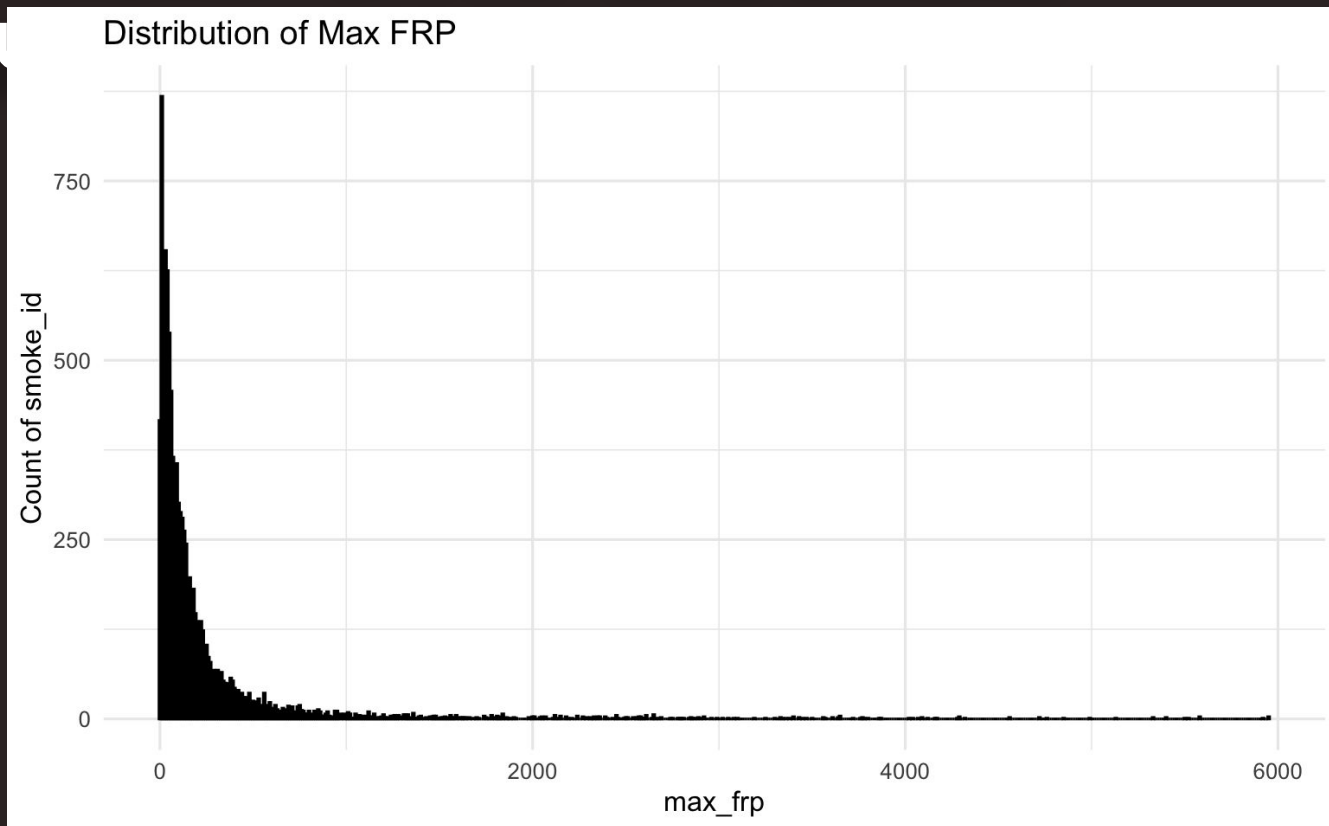


Ecosystems and Fire/Smoke Occurrences



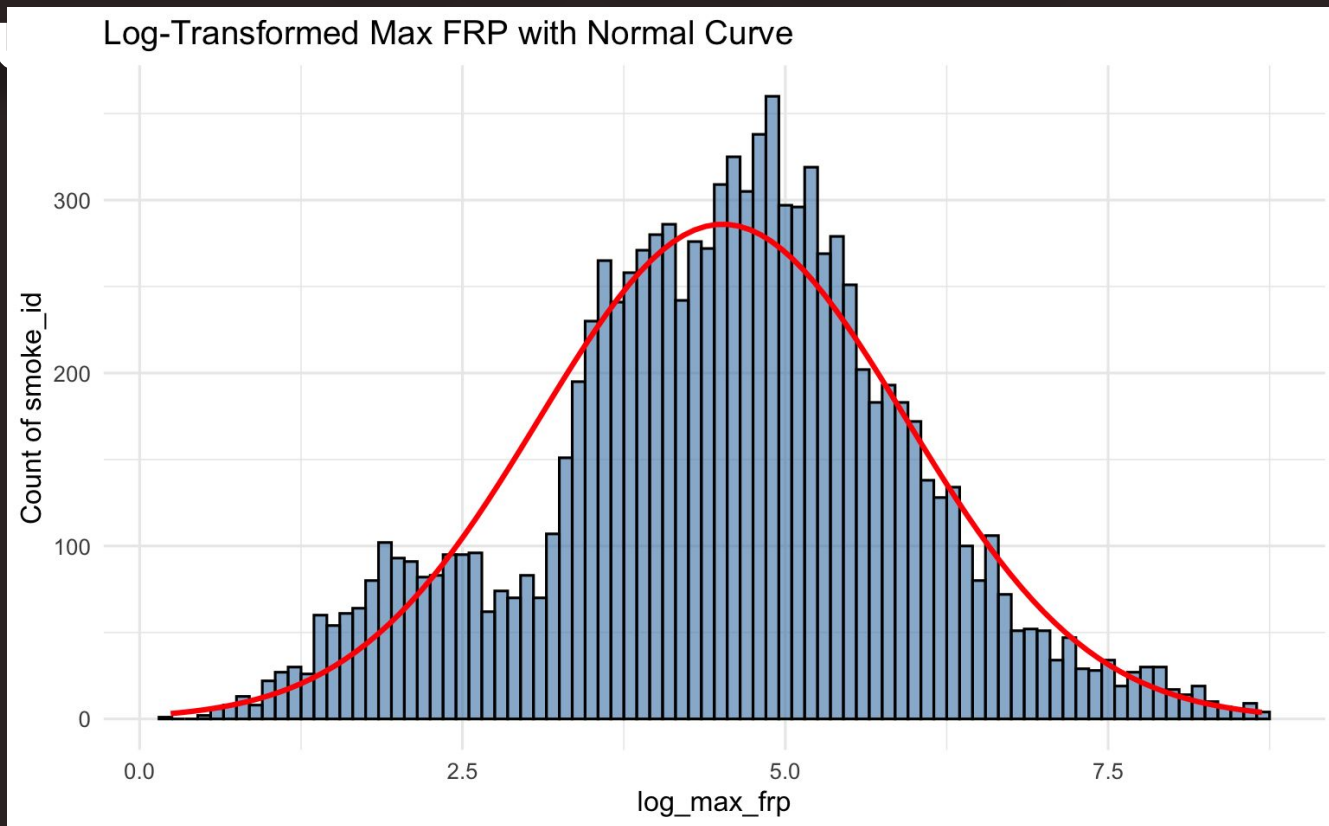
Correlation of FRP and Smoke

Occur



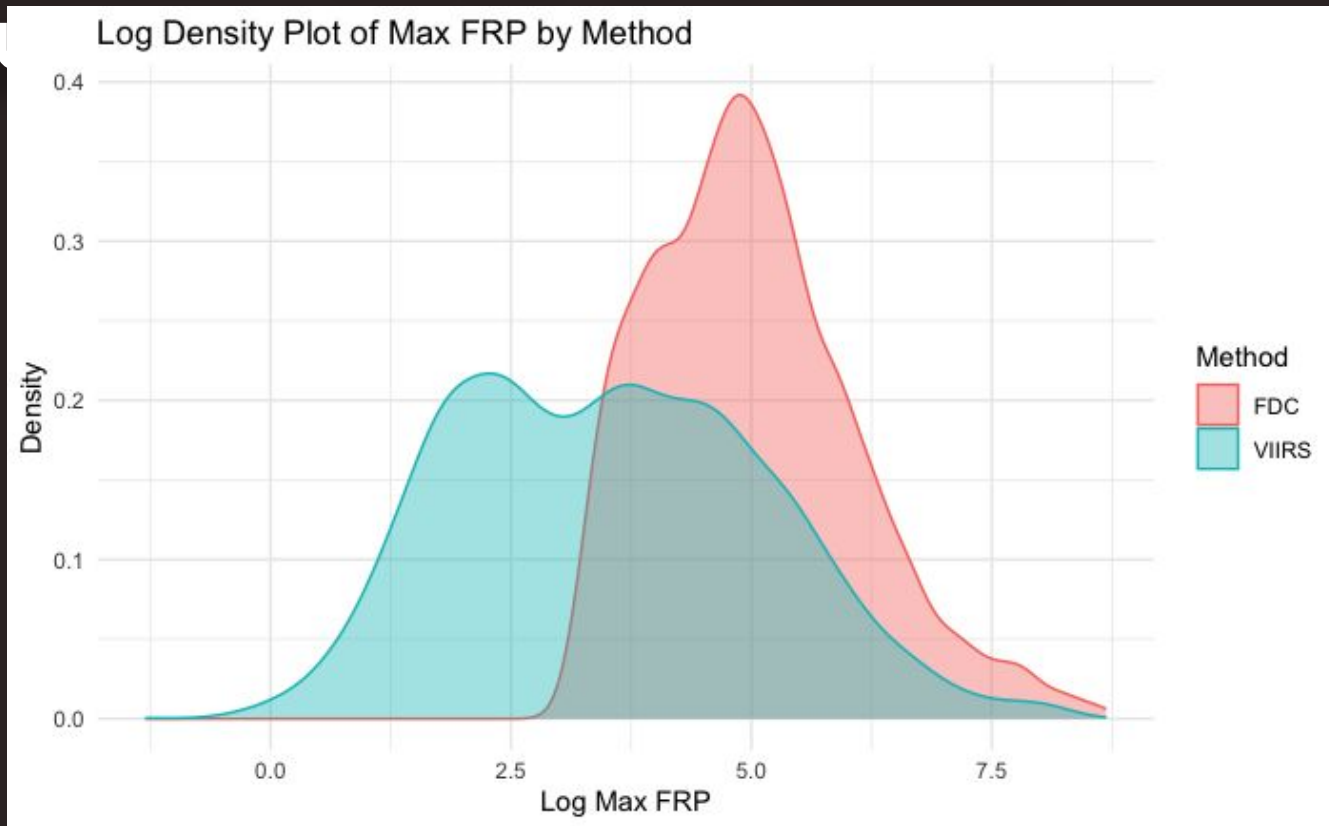
Correlation of FRP and Smoke

Occur

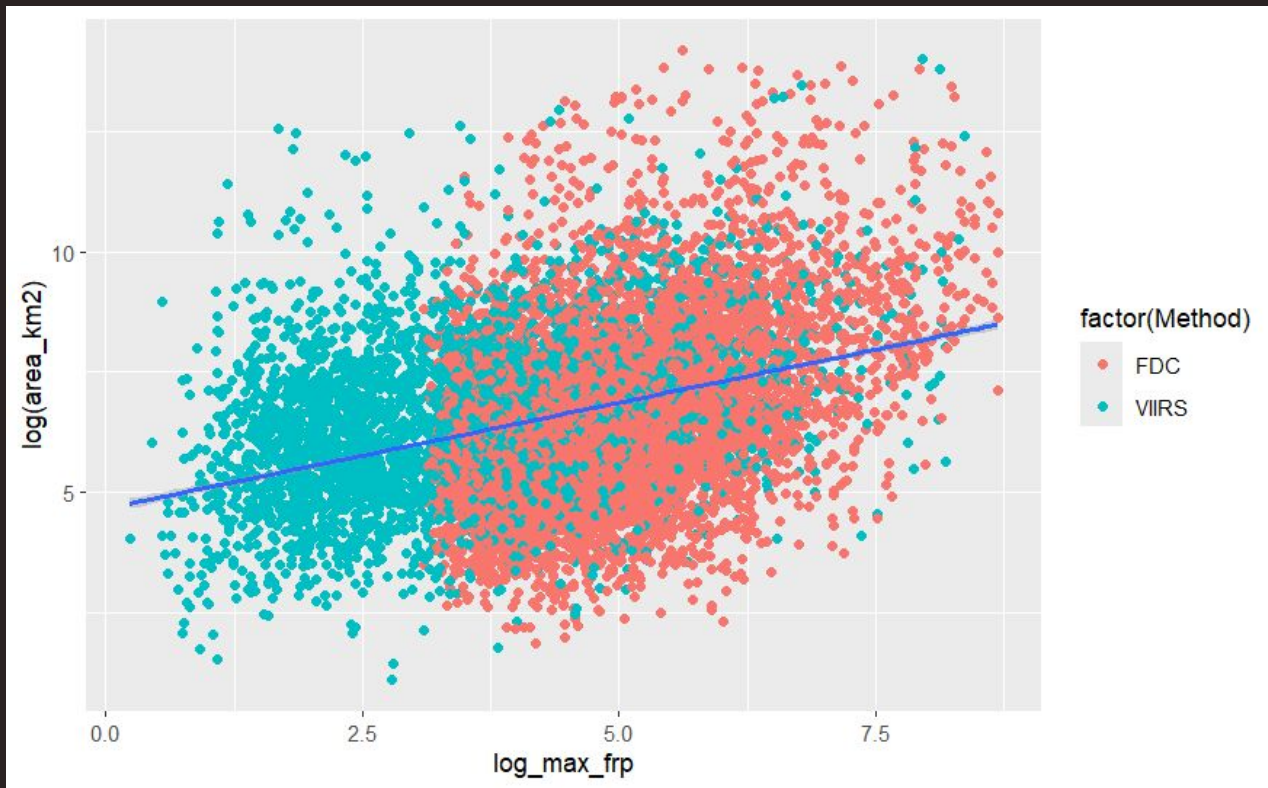


Correlation of FRP and Smoke

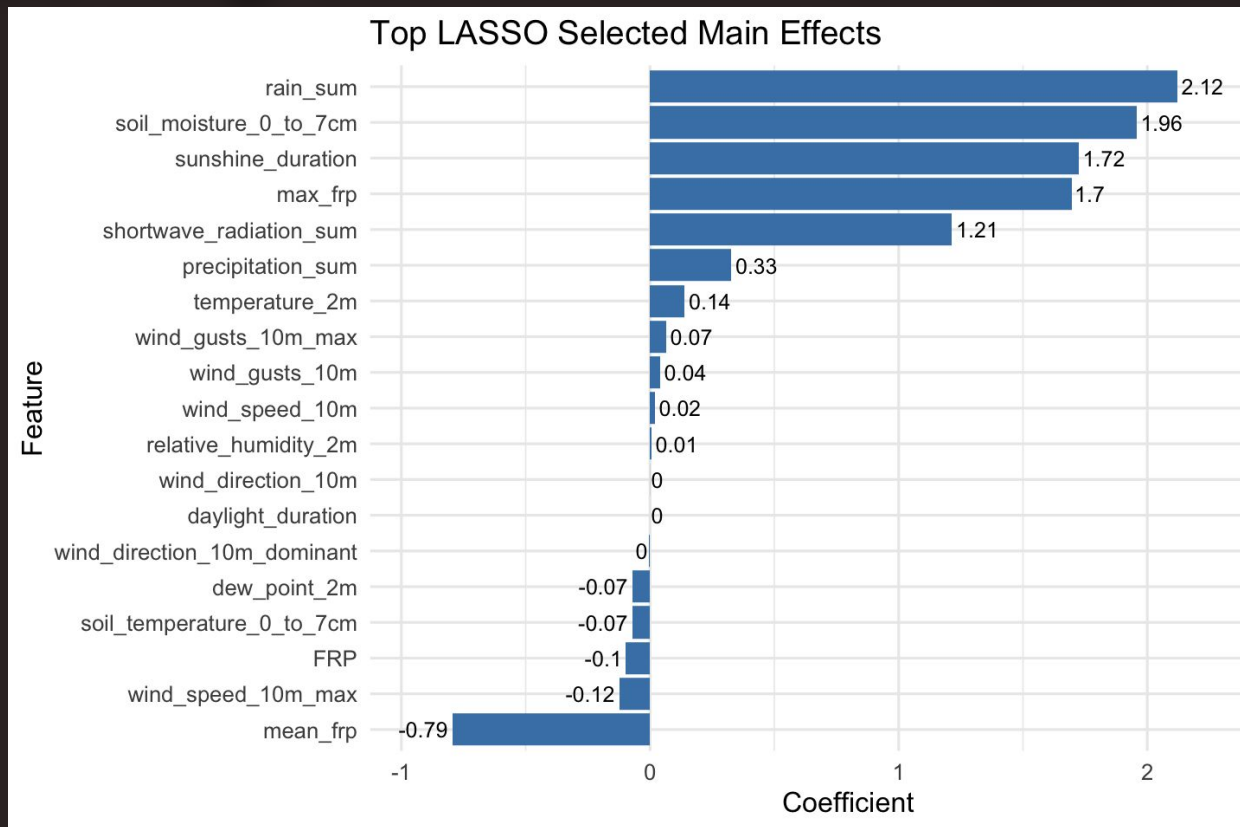
Occu



Correlation of FRP and Smoke Area



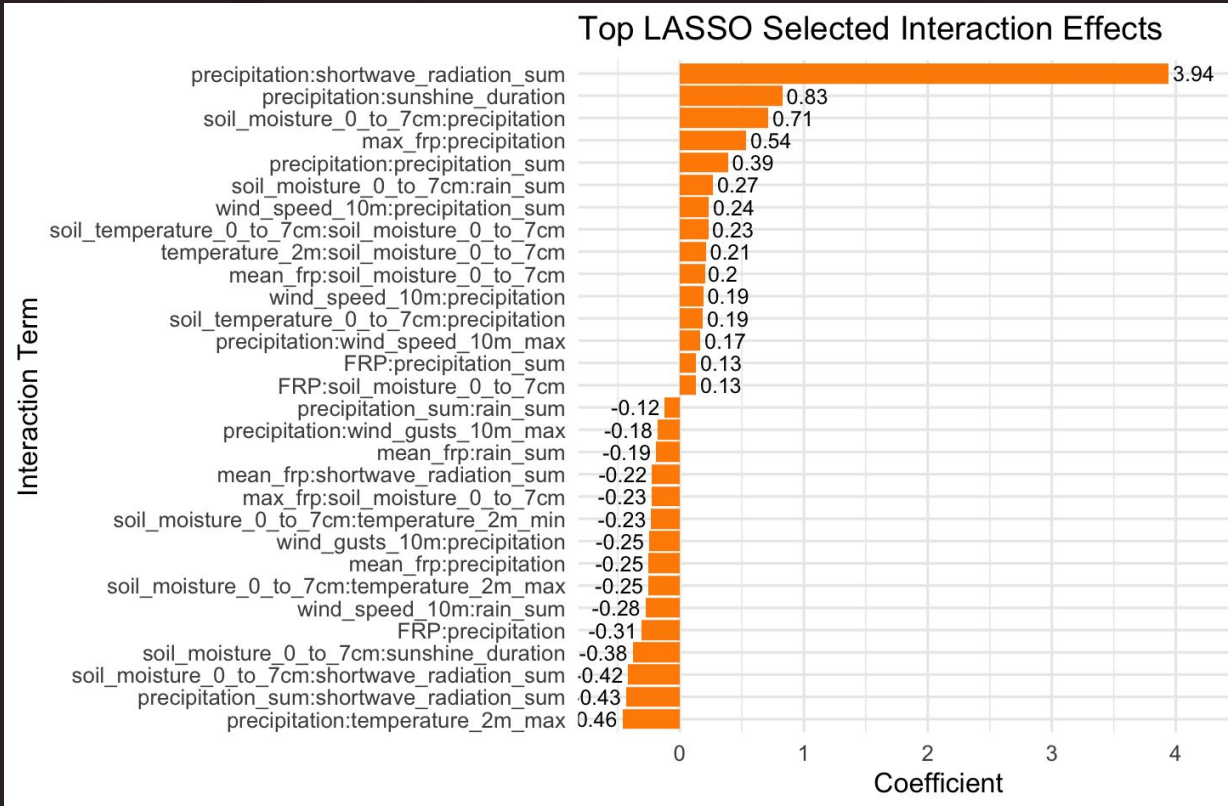
LASSO Regression with Main Effects



RMSE: 1.5093

R-squared: 0.2989

LASSO Regression with Interactions



After Monte Carlo Cross

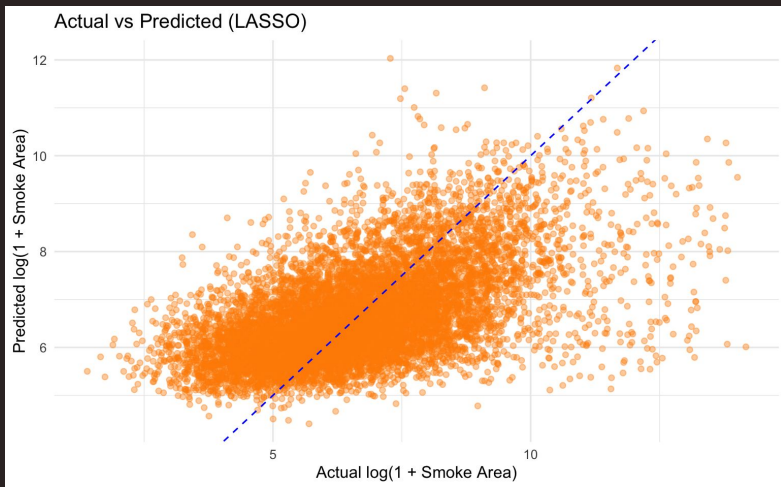
Validation on n=50

repeats:

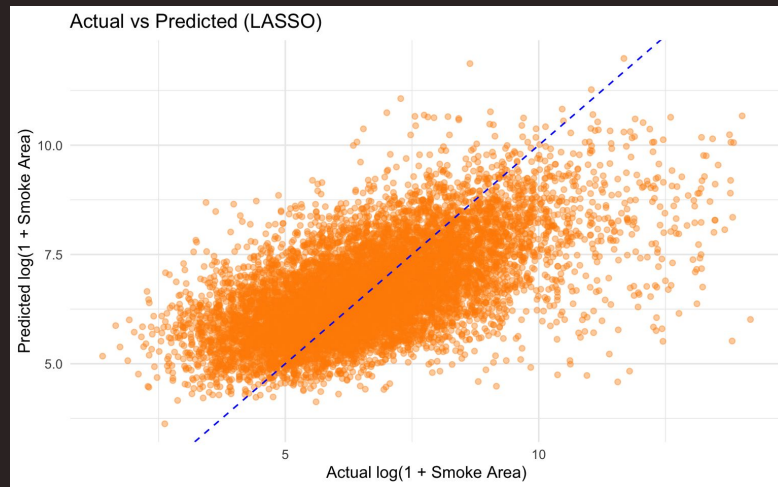
Mean RMSE: 1.465

Mean R-squared: 0.336

Lasso Validation

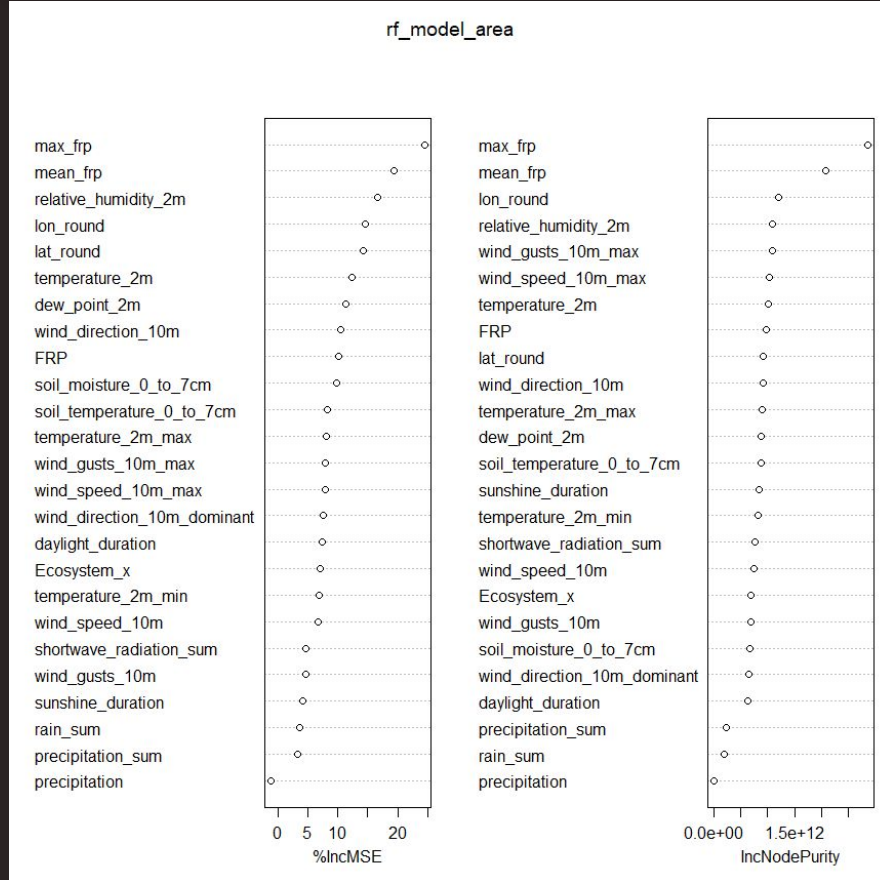


LASSO Regression (Main Effects)



LASSO Regression (Main + Interaction Effects)

Random Forest Analysis



Purpose: To capture nonlinear relationships and rank predictor importance

Target: Area_km2

Predictors: All weather + fire features

Top predictors :

max_frp

mean_frp

relative_humidity_2m

temperature_2m

lon_round / lat_round

Random Forest Analysis

Performance Metrics:

Average Smoke Event: 7,667 km²

RMSE: 47267.78 km²

- Average error in predicted smoke area

R²: 0.1025

- Model explains ~10.25% of variance in area

Milestones

Week 1-3: Data Gathering/Organization & Coding

Week 4-5: Data Analysis of Immediate Effects of Smoke

Week 6-7: Data Analysis of Long Term Effects of Smoke

Week 7: Midterm Presentation

Week 8: Final Outline of Models and Visualization

Week 9: ICS Expo

Week 10: Final Presentation

Thank You!